

Impact of calibration set size for predicting soil fertility attributes using local pXRF spectral libraries

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ABSTRACT

Employing portable Energy-Dispersive X-Ray Fluorescence (pXRF) sensors and Machine Learning (ML) using calibration set sizes with reduced samples can save resources and time in measuring soil fertility attributes. The calibration of local models must have a sufficient number of samples to represent the spectral variation of the area related to the properties of interest, avoiding excess, which would be redundant, and lack, which would cause a loss of performance. From this perspective, this study investigated the impact of different calibration set sizes on the performance of Partial Least Square Regression (PLS) models using a pXRF sensor (0–15 kV) to predict Exchangeable Potassium (K^+), and Calcium (Ca^{2+}), Cation Exchange Capacity (CEC), and Soil Organic Carbon (SOC). Two Brazilian tropical fields composed the pXRF spectral libraries. The Cambé and Toledo datasets comprised 386 and 205 soil samples, respectively. Applying the Kennard-Stone sampling selection algorithm, calibration set sizes ranged from 36 to 276 samples, with increments of 20, for the Cambé field, and from 20 to 140 samples, also with increments of 20, for the Toledo field. Results evaluated on validation sets indicate that calibration sample size can be optimized without compromising model accuracy. Established by quantitatively comparing the models RMSEP, the optimal number of calibration samples was attribute- and field-specific. For the Cambé library, the optimal calibration set size was 116, 136, 156 and 116 for the SOC, CEC, K^+ and Ca^{2+} , respectively. For the Toledo library, the SOC, CEC, K^+ , and Ca^{2+} had optimal sample numbers of 60, 100, 80, and 80, respectively. All these models presented satisfactory RPIQ values (≥ 1.7). Therefore, to encompass all the evaluated attributes, it is suggested 156 calibration samples as a reference for developing local libraries applying pXRF sensors in tropical soils with oxide-rich mineralogy.

1. Introduction

The philosophy of Precision Agriculture (PA) is to manage the spatial and temporal variability in agricultural systems (Gebbers and Adamchuk, 2010a). PA uses technologies to map agronomic attributes and deliver recommendations for variable-rate (VR) applications. This localized management of soil fertility is crucial to ensure the rational use of fertilizers, whilst reducing environmental risks and maximizing crop yields (Nawar et al., 2017). The success of VR fertilizer applications relies on technologies that can rapidly monitor soil fertility status (Mouazen et al., 2020). At the same time, there is a consensus that the accurate mapping of key soil attributes requires high-density sampling

(McBratney et al., 2003). For instance, 50–200 samples per hectare were recommended, depending on the property of interest (Molin and Tavares, 2019).

Collecting samples in the field using a grid sampling design and sending them to soil testing in commercial laboratories is the traditional procedure for mapping soil fertility in PA. In the laboratories, the reference methods for characterizing soil attributes use wet chemistry procedures, which are time-consuming and depend on the use of chemical reagents (José A.M. Demattê et al., 2019; Viscarra Rossel et al., 2016). It may take 3–15 days to analyze dozens of samples, which is not adequate with the short window of time for carrying out agricultural practices in the field, especially PA approaches that demand data

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post-processing (José Alexandre M. Demattê et al., 2019; Gebbers and Adamchuk, 2010b). Furthermore, the traditional procedure for soil mapping also faces prohibitive costs related to the logistics of soil sampling and their subsequent analysis in commercial laboratories. These issues have led to the development of approaches applying soil sensing methods, which are faster, cheaper, and compatible with *in situ* operations by using portable devices (Viscarra Rossel et al., 2011).

Proximal soil sensors (PSS) technologies combined with machine learning (ML) algorithms are an alternative to intensify the spatiotemporal monitoring of soil fertility and mitigate reagent demand (de Santana et al., 2021; Demattê et al., 2017; Ribeiro et al., 2024; Tavares et al., 2021). Recently, portable Energy Dispersive X-ray Fluorescence-based PSS (pXRF) has demonstrated reliable performance for estimating soil attributes (dos Santos et al., 2020; Rawal et al., 2019; Silva et al., 2020, 2017; Tavares et al., 2020a). Portable XRF is a qualitative and quantitative analytical technique for assessing elemental compositions without complex sample preparations. It is founded on the photoelectric effect, i.e., high energy X-rays ionize the samples, the electrons ejected from the inner layer of their atoms leave vacancies that are filled by higher-energy electrons from the outer layers for stabilization. The energy difference between the orbitals involved is manifested in the emission of an electromagnetic wave with specific frequency for each element, called characteristic X-rays (Jenkins, 1988; Van Grieken and Markowicz, 2001). Thereby, pXRF spectra have been widely used to determine the total content of the main soil elements (e.g., Al, Si, P, K, Ca, Mn, Fe, among others). When applied as a soil fertility analysis tool, its spectra are used as a proxy of soil attributes.

To achieve reliable predictions in an agricultural field using pXRF sensors and ML models, a representative dataset (i.e., a local soil spectral library) should be selected on the basis of fertility properties and spectral features (Aichi et al., 2009; Guerrero et al., 2010; Lucà et al., 2017; Nawar and Mouazen, 2018). This is necessary to ensure that the model effectively captures the local relationships between spectra and target soil attributes. Therefore, the calibration of local ML models must have a sufficient number of samples to represent the area, avoiding excess, which would be redundant, and lack, which would cause a loss of performance. On the other hand, it is also essential to optimize costs related to laboratory wet analyses. Consequently, the optimal number of samples required to model calibration with a desired level of accuracy should be established.

The optimization of the number of samples used in local libraries for the calibration of ML models has already been studied when employing spectral libraries of Visible and Near-Infrared Spectroscopy (Vis-NIRS) sensors. Some studies have noted that the number of calibration samples significantly impacts model performance (Brown et al., 2005; Debaene et al., 2014; Gogé et al., 2014; Lucà et al., 2017; Ng et al., 2020; Ramirez-Lopez et al., 2014). With Vis-NIRS, Debaene et al. (Debaene et al., 2014) assessed the performance of Partial Least Square Regression (PLS) models calibrated using different calibration set sizes, ranging from 20 to 200 samples (increments of 20), collected in an agricultural field in Poland. The authors showed that 79 samples are enough for calibrating local models for Soil Organic Carbon (SOC) and soil texture (silt, clay, and sand), demonstrating that the whole dataset (i.e., 200 samples) did not significantly improve the prediction ability. Using a French local library of Vis-NIRS spectra and calibration set sizes varying from 10 to 94 samples (increments of 10), Gogé et al. (Gogé et al., 2014) achieved accurate predictions for clay, Cation Exchange Capacity (CEC), Calcium Carbonate (CaCO_3), and iron (Fe) contents, as well as acceptable predictions for organic carbon by applying PLS models based on 94 calibration samples. Lucà et al. (Lucà et al., 2017) assessed the effect of the calibration set size for predicting total carbon (TC) through a local Vis-NIRS library in Italian soils. Ten calibration subsets ranging from 14 to 144 samples were randomly selected and compared with the full calibration set (144 samples). Prediction results showed that 115 samples were the optimal number of samples required in PLS modeling. Despite several studies involving Vis-NIRS sensors, no study has been

published evaluating the minimum number of calibration samples needed for predicting soil properties using local pXRF libraries.

Therefore, this study was motivated by the lack of scientific information on the optimum number of samples to calibrate pXRF models for predicting key soil fertility attributes in agricultural fields. The objective is to assess the effect the calibration set size for the prediction of exchangeable potassium (K^+) and calcium (Ca^{2+}), CEC and SOC using local soil spectral libraries from two Brazilian agricultural fields and to propose an optimum number of samples for each scenario.

2. Materials and methods

2.1. Study area and reference analyses

Two agricultural fields from a humid subtropical (Cfa) area were explored in this study. The first, designated as Cambé field, is located in the municipality of Cambé, Paraná State, Brazil, and features a local library of 386 soil samples. Cambé field is managed under the no-tillage system with soybean and corn as predominant crops. The soils are basalt-derived and classified as Red Eutrophic Latosol (Rhodic Ferrasol) at the highest altitude (between 544 and 529 m) and Red Eutrophic Nitisol (Rhodic Nitisol) at the lowest altitude (between 529 and 514 m) according to Brazilian Soil Classification System and International Soil Classification System (in parenthesis) (dos Santos et al., 2006; FAO, 2014).

The second field, designated as Toledo, is located in the municipality of Toledo, Paraná State, Brazil, and has a local spectral library of 205 soil samples. This field has been cultivated with grains (also alternating between soybeans and corn) in summer and winter crops in the mid-season. The soils are composed of Red Dystrophic Latosol (Rhodic Ferrasol), also derived from basalt (dos Santos et al., 2006; FAO, 2014). Both agricultural fields exhibit soils in an advanced degree of weathering, leading to elevated levels of iron oxides, aluminum, 1:1 clay minerals (kaolinite), and quartz (Barbosa et al., 2015; Brossard et al., 2012). The main difference between the two fields lies in the greater natural fertility of eutrophic soils compared to dystrophic soils, measured by the Base Saturation Percentage (BS), i.e., $\text{BS} > 50\%$ for eutrophic soils and $\text{BS} < 50\%$ for dystrophic soils (dos Santos et al., 2006). The spectral libraries of both fields included surface (0–10 cm) and subsurface (10–40 cm) samples. The map on Fig. 1 presents a visual detailed description of two experimental fields studied.

After sampling, the soil samples were dried at 40°C for 48 hours, grounded, and sieved through 2 mm mesh, then stored in loose powder form. Reference analyses were carried out in the soil laboratory of the Paraná Rural Development Institute (IDR-PR) to determine the contents of SOC, CEC, Mg^{2+} , K^+ , Ca^{2+} , and potential soil acidity ($\text{H}^+ + \text{Al}^{3+}$). SOC was quantified by dichromate oxidation method; Mg^{2+} and Ca^{2+} were extracted by KCl 1 mol L^{-1} solution and determined by atomic absorption spectrometry, while the K^+ content was extracted using Mehlich-1 solution and determined by flame spectroscopy. The $\text{H}^+ + \text{Al}^{3+}$ was estimated by the SMP extractor, and these results led to the CEC calculation, as indicated by the Eq. 1.

$$\text{CEC} = \text{Mg}^{2+} + \text{K}^+ + \text{Ca}^{2+} + \text{H}^+ + \text{Al}^{3+} \quad (1)$$

2.2. XRF measurement

Portable XRF analyses were performed using a Tracer 5i (Bruker Inc., USA) handheld device, which has an Rh X-ray tube and a Silicon Drift Detector (SDD) with 2048 channels. Approximately 5 g of soil were accommodated in polyethylene pXRF cups (no.1530, Chemplex Industries Inc., USA), which were sealed in the bottom with Mylar film of 2.5 μm thickness (Chemplex Industries Inc., USA). This amount of material resulted in around 10 mm thickness inside the cup, which can be considered an infinitely thick sample for the range of X-ray energy used (Van Grieken and Markowicz, 2001).

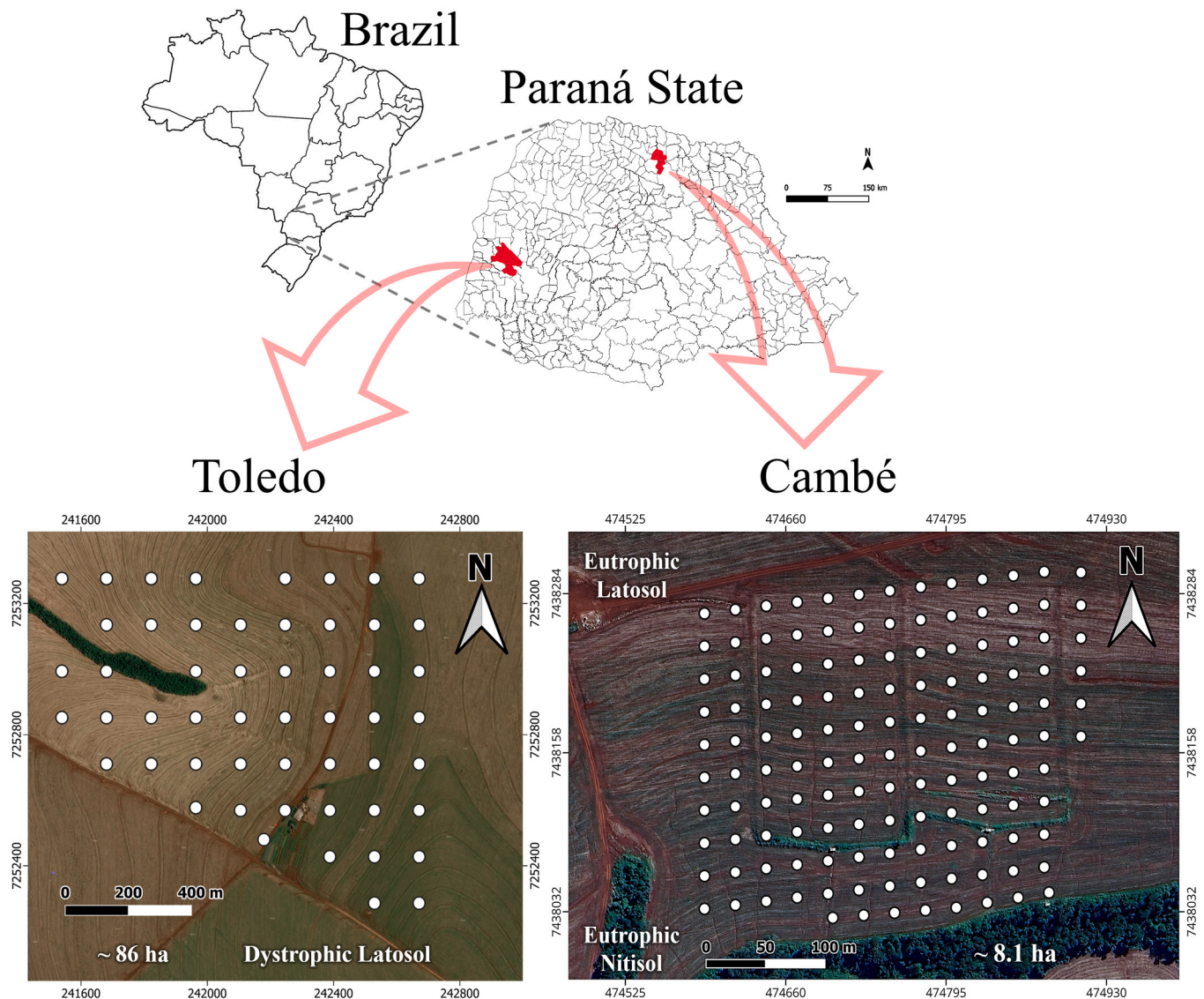


Fig. 1. Location of agricultural fields and sampling point used to compose the local libraries.

The X-ray tube was configured to 15 kV of voltage and 12.15 μA of current, then measurements were made without primary filters and in air atmosphere during 30 seconds for each sample, as suggested by Ribeiro et al. (Ribeiro et al., 2024). The measurements were taken in triplicate, gently stirring the pXRF cup before each measurement. For normalization, the spectra were divided by the tube current and the detector's live time, resulting in intensity unit of photons counts per second per microampere ($\text{cps } \mu\text{A}^{-1}$). Two parts of spectra were removed. The first (0.00–0.98 keV), which presents a standard signal inserted by the equipment manufacturers, while the second (15.02–30.00 keV) includes a background without characteristic emission lines. The final energy range used was 1.0–15.0 keV. For the subsequent analysis, the average spectra of each replicate were calculated.

2.3. Sample selection and calibration strategies

Kennard-Stone (KS) (Kennard and Stone, 1969) was used as the sample selection algorithm. This approach has been applied in several chemometrics and soil spectroscopy studies, and it has proven to be effective in representatively splitting samples into calibration and prediction (or validation) sets (dos Santos et al., 2020; Nawar and Mouazen, 2018; Ribeiro et al., 2024). Specifically, KS ensures a uniform

distribution of the subset samples over the entire dataset by following a stepwise procedure in which the Euclidian distance of each pair of samples is employed (Kennard and Stone, 1969; Nawar and Mouazen, 2018).

In the present study, KS was firstly applied on the fertility attributes matrix to obtain the prediction dataset (approximately 30 % of the samples), and then used on the remaining samples to obtain the calibration sets with reduced sizes. For instance, the 386 samples of Cambé were split into 276 (full calibration) and 110 (prediction) sets. Then, the full calibration set was reduced from 276 to 36 samples with a decrement of 20 samples. In this process, 13 calibration sets were generated, comprising 276, 256, 236, 216, 196, 176, 156, 136, 116, 96, 76, 56, and 36 samples. Similarly, the 205 samples of the Toledo experimental field were split into 140 (full calibration) and 65 (prediction) sets. Then, the full calibration set was also reduced, from 140 to 20 with a decrement of 20 samples. In this process, 7 calibration sets were generated, which comprised 140, 120, 100, 80, 60, 40, and 20 samples. Finally, models using all different calibration set sizes were built and the prediction sets were used to check the model's performance in each scenario.

2.4. Predictive modelling and evaluation metrics

Before data modeling, pXRF data were pre-processed using Poisson scaling and mean centering, as suggested by Dos Santos *et al.* (dos Santos *et al.*, 2021b). Poisson pre-processing normalizes the data by considering its individual uncertainty (Keenan and Kotula, 2004; Van Grieken and Markowicz, 2001). As a result, emission peaks of lower magnitudes (typically originating from elements with lower contents such as P, S, and K) are evidenced.

The preprocessed spectral data of each calibration set size scenario were used to build PLS models for quantifying SOC, CEC, Ca^{2+} , and K^+ . PLS was chosen as it is one of the most popular and efficient methods for modeling local spectral libraries (dos Santos *et al.*, 2020; Ribeiro *et al.*, 2024; Tavares *et al.*, 2021). In addition to being efficient for local databases, it is a simple ML algorithm compared to others (e.g., Support Vector Machine, Random Forest, Artificial Neural Networks, among others). PLS models the relationship between the $\mathbf{X}$ -predictor variables (e.g., XRF spectra) and the $\mathbf{y}$ -target vector (e.g., a soil fertility attribute) by building a mutual space defined by latent variables (LVs) that capture shared information between them (Geladi and Kowalski, 1986; Helland, 1990). Each LV represents a unique contribution to this relationship, and tuning its number is crucial to balance model complexity and predictive accuracy. In this study, the LVs of each calibration set scenario were defined through 10-fold cross-validation, seeking to minimize the root mean square error (RMSE) of cross-validation (RMSECV).

Furthermore, PLS also has the advantage of being easy to interpret, *e.g.*, the most important predictor variables can be identified through different parameters, including the Variable Influence on Projection (VIP) scores. Specifically, VIP scores uncover the impact of individual $\mathbf{X}$ -predictors on the model by considering a weighted sum of squares of the PLS weights ($\mathbf{W}$), which incorporate the proportion of $\mathbf{y}$ -variance explained by each LV (Farrés *et al.*, 2015; Wold *et al.*, 1993). Mathematically, the VIP score for the i -th $\mathbf{X}$ -variable is computed according to Eq. 2,

$$VIP_i = \sqrt{\frac{\sum_{l=1}^{LV} W_{il}^2 \times SSY_l \times K}{LV \times SSY_{total}}} \quad (2)$$

in which LV is the total number of PLS components, W_{il} is the i -th weight for the l -th LV, SSY_l is the sum of squares of explained variance for the l -th LV, SSY_{total} is the total sum of squares explained of the $\mathbf{y}$ -vector, and K is the total number of $\mathbf{X}$ -variables. Since the mean of the squared VIP scores is 1, "greater than unity" is used as a criterion to establish the significant variables for modeling (Farrés *et al.*, 2015).

The model's performance was evaluated through the RMSE of prediction (RMSEP), coefficient of determination (R^2) and the ratio of performance to interquartile distance (RPIQ), as recommended by Bellon-Maurel *et al.* (Bellon-Maurel *et al.*, 2010) and defined as:

$$RPIQ = \frac{IQ}{RMSEP} \quad (3)$$

where:

$$RMSEP = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (4)$$

and IQ is the difference between the third and first quartiles of $\mathbf{y}$ ($IQ = Q_3 - Q_1$); y_i and $\hat{y}_i$ are the observed and predicted value, respectively; n is the number of samples with $i = 1, 2, \dots, n$.

2.5. Accuracy comparison test

RMSE, R^2 and RPIQ are valuable metrics of model performances. Nonetheless, a proper assessment of the equivalence between two ML

models should not be founded on visual comparison of these indicators (Olivieri, 2015). A statistical test was used to evaluate the optimal calibration set sizes required to derive PLS models with accuracy equivalent to those calibrated with the full calibration set. Proposed by Van der Voet to compare prediction errors between multivariate methods, the Randomization Test quantitatively evaluates when the RMSEP values of two compared models are statistically different (van der Voet, 1994). This approach starts by computing the mean difference between the prediction square errors between such models. Subsequently, each prediction square error difference is randomly inverted (*i.e.* randomly multiplied by -1) and the resulting new mean is compared to the original one. This process is repeated j -times ($j = 1999$ in this study) while a statistic on the relative number of times the new mean error difference is larger than the original one is registered. Afterward, the associated p -value undergoes a hypothesis test, where the significance of the differences between the original and inverted mean errors are assessed. For a 95 % level of confidence, p -values > 0.05 (null hypothesis) indicate that the mean errors are equivalent, *i.e.*, the models' predictive accuracy can be considered similar (in RMSEP terms). The p -values < 0.05 (alternative hypothesis) correspond to scenarios where this predictive accuracy is not equivalent (Olivieri, 2015; van der Voet, 1994).

All pre-processing and computational analyses were performed in the R Statistical Software (v4.3.3, (R Core Team, 2024)). The dplyr package (v1.1.4, (Wickham *et al.*, 2023)) was used for data manipulation, prospectr (v0.2.7, (Stevens and Ramirez-Lopez, 2024)) was used to split the samples by the Kennard-Stone as well as mdatools (v0.14.1, (Kucheryavskiy, 2020)) was used for PLS modeling.

3. Results and discussion

3.1. Conventional analyses

Table 1 shows the descriptive statistics of the reference analyses. The mean levels of SOC, K^+ , and Ca^{2+} for both fields can be considered high compared to the reference for tropical soils such as in the state of Paraná in Brazil (Pavinato *et al.*, 2017). As these soils present low natural fertility, such K^+ and Ca^{2+} levels (which ranged from 0.02 to 1.3 and 0.4–9.2 $\text{cmol}_c \text{ kg}^{-1}$, respectively) reflect the applications of lime and fertilizers. SOC represents the amount of carbon present in organic matter, which is generally associated with the soil clay (Prezotti and Guarçoni, 2013). Thus, the mean SOC levels found in both fields (which ranged from 8 to 34 g kg^{-1}) are typically from soils with clayey texture conditions (Cantarella *et al.*, 2022). Moreover, both fields have CEC values between 7.0 and 17.9 $\text{cmol}_c \text{ kg}^{-1}$, typical levels for tropical soils with mineralogy dominated by oxides and kaolinite (Hartemink, 2002).

On the other hand, the CEC value of the Cambé field presented a coefficient of variation (CV) smaller than 12 %, while in the Toledo field, it showed a CV of 19 %. SOC and Ca^{2+} of both fields presented a medium variability ($12 \% < CV < 60 \%$) while CV values for K^+ were the high ones ($> 60 \%$) (Warrick and Nielsen, 1980). The spatiotemporal variation of fertility attributes in agricultural soils is mainly related to anthropogenic factors (Hartemink, 2002), such as the application of fertilizers and amendments, and subsoil exposure when raising terraces. Other factors related to a heterogeneous export of nutrients (*i.e.*, due to pest or disease attacks) and abrupt natural changes in soil formation factors can also lead to this variability in local databases (Molin and Tavares, 2019).

3.2. pXRF spectral data

Fig. 2 shows the mean pXRF spectra according to the two agricultural fields. The instrument conditions allowed the identification of Al, Si, P, S, K, Ca, Ti, Mn, Fe, Cu and Zn, in which the Fe and Ti K-lines presented the highest intensities. The Rh scattering L-lines from the tube target are also identified, however, they overlap the Ar $K\alpha$ -line (2.958 keV),

Table 1
Descriptive statistics of the reference analyses for the two studied fields (Cambé *n* = 386, Toledo *n* = 205).

Attribute	SOC (g kg ⁻¹)		CEC (cmol _c kg ⁻¹)		K ⁺ (cmol _c kg ⁻¹)		Ca ²⁺ (cmol _c kg ⁻¹)	
Field	Cambé	Toledo	Cambé	Toledo	Cambé	Toledo	Cambé	Toledo
Mean + SD	20 ± 4	17 ± 6	13.7 ± 1.5	11 ± 2	0.24 ± 0.16	0.3 ± 0.2	6.0 ± 1.5	3.6 ± 1.8
CV (%)	22	34	11.3	19	66.78	80.8	25.2	48.9
Median	20	17	13.8	11	0.18	0.2	6.2	3.4
Minimum	8	4	9.3	7	0.05	0.02	1.4	0.4
Maximum	31	34	17.9	16	1.15	1.3	9.2	9.1
Kurtosis	−0.5	−0.2	−0.3	−0.27	3.53	1.2	−0.5	−0.2
Skewness	−0.3	0.2	−0.2	0.3	1.57	1.1	−0.4	0.6

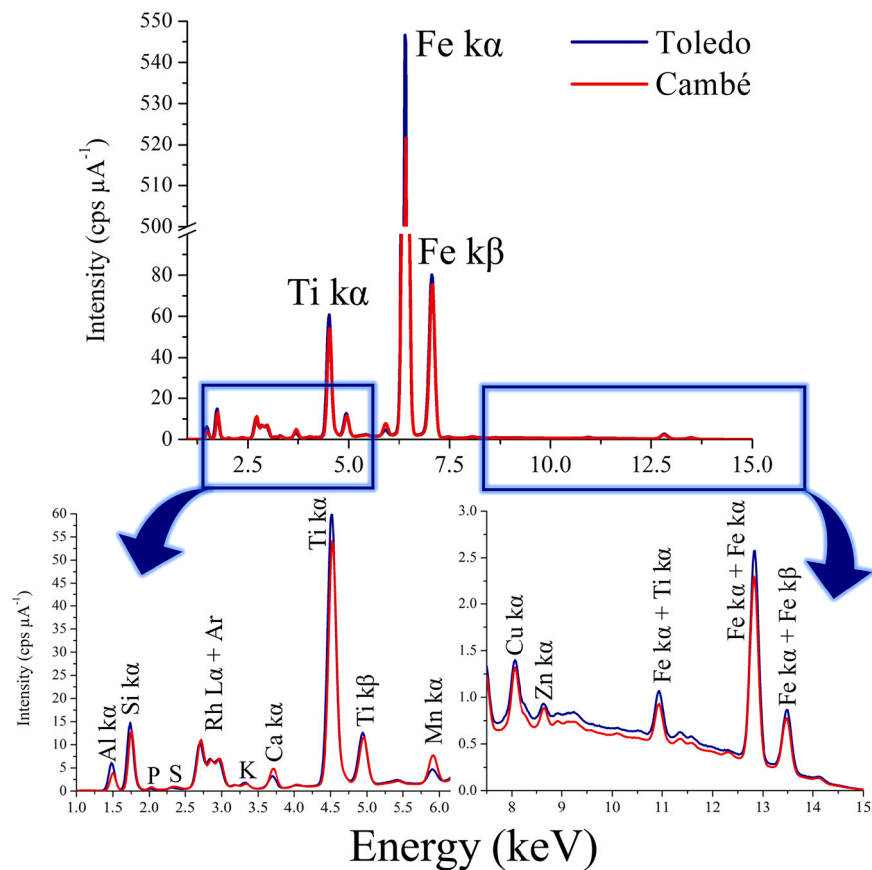


Fig. 2. Mean pXRF spectra of the two agricultural fields (Cambé *n*=386, Toledo *n*=205).

caused by the fluorescence of this element that is present in the atmospheric air between the sample and the detector.

The characteristic X-ray patterns for both fields are qualitatively similar, although some particularities regarding their mean intensity may be noted. Toledo has a higher average emission of Fe- α , Ti- α , and Al- α than Cambé, which should be related to the greater presence of oxides in Toledo, such as hematite (α -Fe₂O₃), goethite (FeOOH), and gibbsite [Al(OH)₃], which have soils classified as Red Dystrophic Latosol. Eventually, Ti can also be seen in the iron oxides structure as isomorphous substitutes for Fe (Singh and Gilkes, 1992), explaining the greater presence of Ti in soils rich in oxides. In addition, the dystrophic nature of Toledo (which indicates low base saturation, e.g. Ca²⁺, Mg²⁺, K⁺) compared to the eutrophic nature of Cambé corroborates with the spectral behavior showing a higher emission of Ca in Cambé in relation to Toledo. Dystrophic Latosols are very old and well-developed soils that have undergone a long period of weathering and leaching, resulting in nutrient-poor soils with a simplistic mineralogy.

3.3. Correlations between fertility attributes and pXRF spectra

Fig. 3 shows the Pearson's correlogram between the soil fertility attributes by reference methods and pXRF spectra according the two agricultural fields. The Pearson's correlation coefficients (*r*) were interpreted as weak (0.00 < *r* < 0.29), moderate (0.30 < *r* < 0.59), and strong (*r* > 0.60) (Asuero et al., 2006; Ege and Kamali, 2024). It is noticed that the P, S, K, and Ca emission k-lines are the pXRF variables most correlated with all fertility attributes, simultaneously (*r* ranging from 0.22 to 0.63 for P, 0.10–0.62 for S, 0.33–0.82 for K and 0.36–0.83 for Ca). For Cambé samples, SOC had strong correlations with the P, S, and Ca emission k-lines (*r* > 0.60), moderate correlations with Si and K (0.30 < *r* < 0.59), and weak correlations with all remaining pXRF variables (*r* < 0.29). CEC had strong correlations with K and the Ca k-line, moderate correlations with Al, P, Mn, and Cu, and a weak correlation with the remaining pXRF variables. K⁺ had strong and moderate correlations with K and Ca emission lines, respectively, presenting weak correlations with the other pXRF variables. Ca²⁺ had strong correlations with Ca and K emission k-lines, moderate

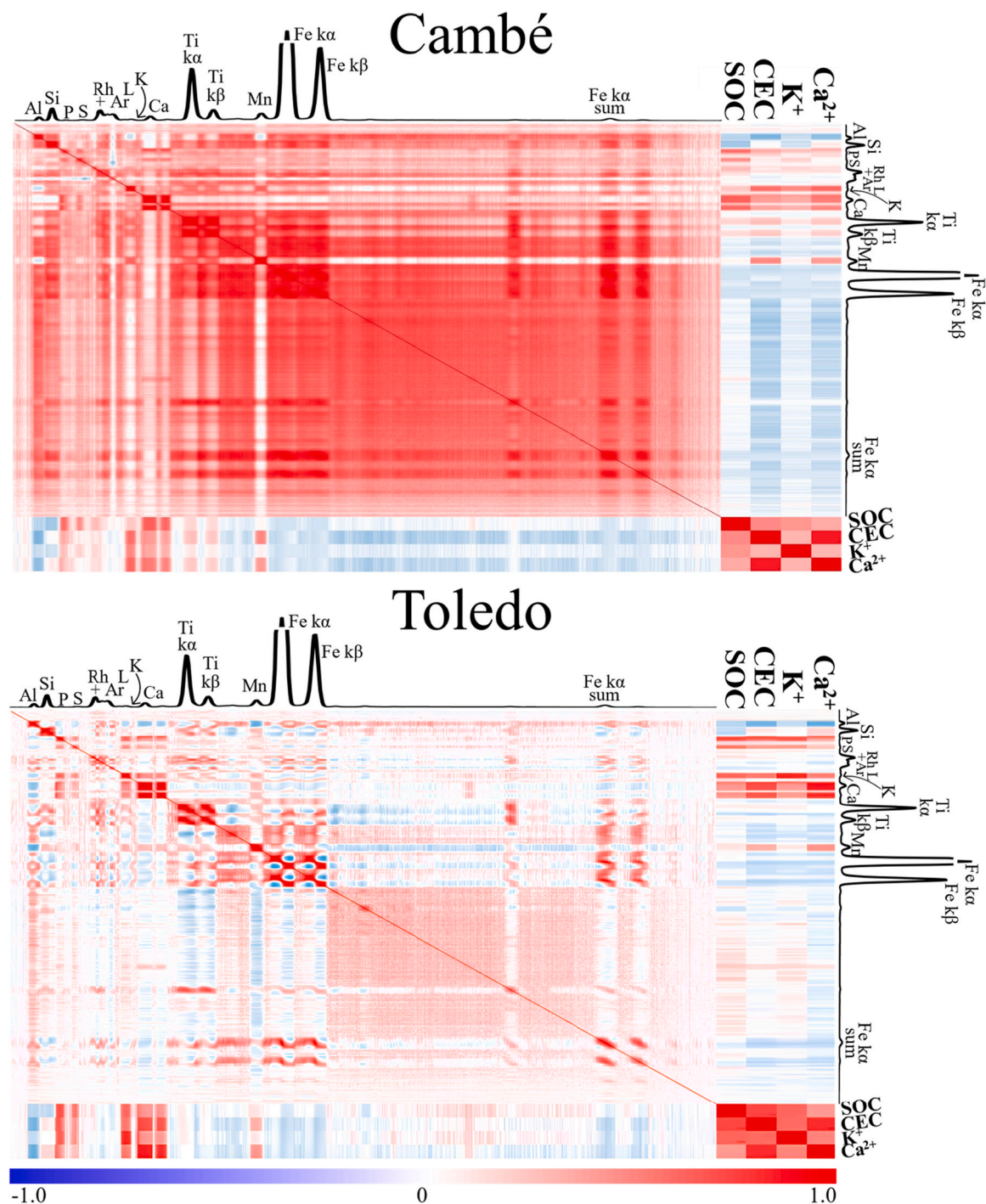


Fig. 3. Pearson correlogram of the soil fertility attributes and XRF spectra obtained using the entire dataset of each local spectral library.

correlations with Al, Mn, and Cu, as well as weak correlations with the other pXRF emission lines. For the Toledo dataset, SOC had strong correlations with the P, S, K, and Ca emission k-lines while showing weak correlations with all remaining pXRF variables. CEC had strong correlations with P, K and Ca, moderate correlations with Al, S, Mn, and Fe in addition to weak correlations with the other pXRF variables. K^+ had strong correlations with P and K, moderate with S and Ca in addition to its weak correlation with all the others pXRF lines. Finally, Ca^{2+} had strong correlations with K and Ca, moderate with Al, P and Mn, along with a weak correlation with the remaining pXRF variables. Furthermore, the samples from Toledo showed correlation intensities between pXRF variables and soil attributes higher than the Cambé. These patterns

favor modeling with Toledo samples, *i.e.* data with higher correlations are better for modeling as the stronger relationship between variables is explored to relate them quantitatively.

Evaluating the interrelationships between soil attributes in different areas, their mutual correlations ranged from weak ($r=0.30$) to strong ($r=0.86$). For Cambé samples, SOC and K^+ had moderate and weak correlations with the other attributes, respectively. CEC and Ca^{2+} had strong correlations between them, and weak correlations with the other attributes. For Toledo samples, CEC had a strong correlation with all soil attributes while SOC had a strong correlation with K^+ and a weak correlation with Ca^{2+} . K^+ had a weak correlation with Ca^{2+} . The details of the r coefficients are presented in the Supplementary Materials

(Table S.1).

3.4. Predictive performance of different calibration set size scenarios

Table 2 and Table 3 present the performance of the R^2 , RPIQ and RMSEP using the different calibration set size scenarios for both evaluated fields. In general, good predictions ($R^2 \geq 0.73$ and $RPIQ \geq 2.31$) were achieved for most models that used the full calibration set. With the only exception for the prediction of K^+ in Cambé that had R^2 of 0.64 and RPIQ of 1.94. Some attributes decreased their predictive performance more abruptly when the number of samples is reduced compared to others. For instance, the predictive performance of Ca^{2+} always had an RPIQ larger than 2, even with the lower number of samples in the Toledo ($n=20$) and in Cambé ($n=36$) areas. A similar pattern was observed for SOC, which maintained RPIQ values close to 2 even for models calibrated with 40 (Toledo) and 36 samples (Cambé). Conversely, CEC predictions in Toledo had an RPIQ performance of 2.89 when using the full calibration set ($n = 140$) and it dropped to 1.12 when using both 40 and 20 samples for model calibrations. Similarly, K^+ prediction in Toledo reached an RPIQ of 3.80 with the full calibration set, which reduced to 1.70 when using calibration 40 samples. These patterns suggest that the optimum number of samples in the calibration set is attribute-specific. Moreover, it is worth mentioning that 5 out of 8 models (K^+ and Ca^{2+} in Toledo, in addition to SOC, CEC, and Ca^{2+} in Cambé) still showed satisfactory predictive performances with $RPIQ > 1.70$ even with the least number of samples in the calibration set ($n=20$ samples in Toledo and $n= 36$ in Cambé).

Comparing the attributes between the two soil libraries, Toledo presented CEC, Ca^{2+} , and K^+ models with a slightly better performance than Cambé (Tables 2 and 3). This result can be explained as the correlation intensities between the pXRF variables and the fertility attributes from Toledo dataset were greater than the ones from Cambé (see Fig. 3). In contrast, the eutrophic soils of Cambé exhibit a greater variety of clay minerals, compared to Toledo dystrophic soils (Schaefer et al., 2008). This data diversity increases the CEC (Table 1), i.e., enables the soil to retain and exchange a wider range of nutrients, resulting in a more dynamic and complex system of interactions between nutrients, minerals, and organic matter, which can be reflected in modeling. Consequently, the low natural fertility of the dystrophic character associated with the low mineralogical components of Toledo soils may explain its superior predictive performance.

Regarding individual fields, Cambé dataset had SOC models presenting the highest performance (R^2 ranging from 0.76 to 0.46), CEC and Ca^{2+} with similar prediction results (R^2 oscillating from 0.71 to 0.54 and from 0.71 to 0.56 for CEC and Ca^{2+} , respectively) while the K^+ had the lowest performance among all models (R^2 varying from 0.64 to 0.49) (Table 2). For the Toledo dataset, Ca^{2+} had the highest R^2 (ranging from 0.89 to 0.65), while the SOC had the lowest (R^2 oscillating from 0.73 to 0.35) compared with CEC and K^+ models (Table 3). It is also noted that the R^2 values decrease in a non-linear trend as the number samples was reduced.

To better observe the performance as the calibration set scenarios

reduced their number of samples, RMSEP and RPIQ values are visually shown in Fig. 4. Similar to R^2 , RPIQ values decreased, in a non-linear-trend, when decreasing the number of samples in the calibration set. Conversely, RMSEP values increased, non-linearly, when the number of samples in the calibration set diminished. For Toledo, the reduction from 140 to 20 samples in the CEC model had the largest RMSEP increase, from 0.91 to 2.36 $cmol_c\ kg^{-1}$ (159 % increase). For Cambé, the greatest RMSEP increase was observed for SOC, which ranged from 1.85 to 2.75 $g\ kg^{-1}$ (49 % increase) when reducing the calibration set size from 276 to 36 samples. This expected decrease in performance when fewer samples are used was observed in most calibration scenarios. However, as it is shown in Fig. 4, there were some cases with distinct characteristics, i.e., some models calibrated with a slightly smaller number of samples showed equal or better performance than the superior models. This non-linear trend may be attributed to the complex relation between the multicollinearity and redundancy of the information contained in the set of calibration samples. PLS is a machine learning algorithm designed to handle high-dimensional multicollinear data by seeking latent variables that maximize the covariance between the X -predictor and the y -target variable (Geladi and Kowalski, 1986). In large datasets, such as those evaluated in this study, the correlation between variables may introduce redundancies that do not add value to the model, causing a larger number of samples to increase complexity and lead to overfitting. In these cases, reducing the number of samples may simplify the model, eliminating redundant samples and focusing on the most relevant patterns. Beyond the Kennard-Stone algorithm, other similarity-based selection approaches, such as Similarity Analysis and K-Nearest Neighbors, also needed to be tested, as they can be more efficient and allow for even greater reductions without compromising performance (Nawar and Mouazen, 2018)

3.5. Equivalence between the different calibration set sizes and the full calibration set

Table 4 and Table 5 showed the Random Test p -values comparing the results of models calibrated with the full and the reduced calibration set size scenarios for Cambé and Toledo fields, respectively. The criteria of $RPIQ > 1.7$ and Random Test p -value greater than 0.05 (95 % confidence level) were adopted to determine the optimum reduced sample set for each attribute. Such criteria were used as guidance to attain models with adequate performance for quantitative analysis in terms of RPIQ (Nawar and Mouazen, 2017) and RMSEP values equivalent to models calibrated with the full calibration set.

For the Cambé dataset, the minimum number of calibration samples ranged from 156 to 116 (Table 4). SOC and Ca^{2+} prediction reached an optimal calibration set size using 116 samples, while for CEC and K^+ the same was reached with 136 and 156 samples, respectively. For the Toledo dataset, the optimal number of samples ranged from 60 to 100, being 60 for SOC, 100 for CEC, and 80 for K^+ and Ca^{2+} (Table 5). It was noted that soil attributes accurately quantified ($RPIQ > 1.9$) showed better stabilities to reduce the samples in higher numbers. For example, the Toledo SOC model ($RPIQ = 1.94$) presented the smallest number of

Table 2

Performance of the coefficient of determination (R^2), ratio of performance to interquartile distance (RPIQ) and the root mean square error of prediction (RMSEP) obtained for the validation set ($n = 65$) using the different calibration set size scenarios with Toledo spectral library. Full detailed results are presented in the supplementary materials.

Calibration set size	SOC	CEC	K^+	Ca^{2+}	SOC	CEC	K^+	Ca^{2+}	SOC	CEC	K^+	Ca^{2+}
	R^2											
					RPIQ				RMSEP			
140	0.73	0.76	0.80	0.89	2.21	2.89	3.80	4.65	2.58	0.91	0.09	0.58
120	0.70	0.74	0.78	0.89	2.09	2.71	3.62	4.65	2.73	0.95	0.10	0.58
100	0.70	0.69	0.75	0.88	2.16	2.28	3.34	4.56	2.73	0.96	0.11	0.59
80	0.67	0.63	0.70	0.87	1.99	2.33	3.10	4.36	2.86	1.13	0.12	0.62
60	0.65	0.43	0.59	0.85	1.94	1.56	2.65	4.02	2.92	1.41	0.14	0.67
40	0.52	0.07	0.38	0.83	2.05	1.12	1.70	3.17	3.42	2.36	0.17	0.71
20	0.35	0.07	0.23	0.65	1.43	1.12	1.93	2.66	3.97	2.36	0.19	1.02

Table 3
Performance of the coefficient of determination (R^2), ratio of performance to interquartile distance (RPIQ), and the root mean square error of prediction (RMSEP) obtained for the validation set ($n = 110$) using the different calibration set size scenarios with Cambé spectral library. Full detailed results are presented in the supplementary materials.

Calibration set size	SOC	CEC	K ⁺	Ca ²⁺	SOC	CEC	K ⁺	Ca ²⁺	SOC	CEC	K ⁺	Ca ²⁺
	R²				RPIQ				RMSEP			
276	0.76	0.71	0.64	0.71	3.11	2.67	1.94	2.59	1.85	0.75	0.09	0.80
256	0.76	0.71	0.64	0.71	3.10	2.67	1.94	2.59	1.86	0.75	0.09	0.80
236	0.76	0.71	0.62	0.71	3.11	2.66	1.92	2.56	1.87	0.75	0.09	0.80
216	0.76	0.71	0.63	0.71	3.07	2.61	1.92	2.56	1.88	0.76	0.09	0.80
196	0.75	0.7	0.62	0.71	3.09	2.62	1.90	2.56	1.88	0.77	0.09	0.80
176	0.75	0.69	0.63	0.70	3.01	2.62	1.84	2.55	1.91	0.78	0.09	0.81
156	0.74	0.69	0.59	0.69	2.98	2.62	1.83	2.53	1.95	0.79	0.09	0.82
136	0.73	0.68	0.59	0.69	2.95	2.53	1.82	2.56	1.97	0.79	0.09	0.83
116	0.72	0.66	0.59	0.69	2.87	2.47	1.82	2.51	2.02	0.81	0.09	0.83
96	0.71	0.62	0.56	0.67	2.83	2.34	1.76	2.43	2.06	0.86	0.10	0.86
76	0.67	0.61	0.56	0.65	2.67	2.31	1.77	2.36	2.17	0.87	0.10	0.88
56	0.69	0.57	0.50	0.64	2.74	2.20	1.65	2.33	2.10	0.91	0.10	0.89
36	0.47	0.54	0.49	0.56	2.09	2.12	1.63	2.12	2.75	0.94	0.10	0.98

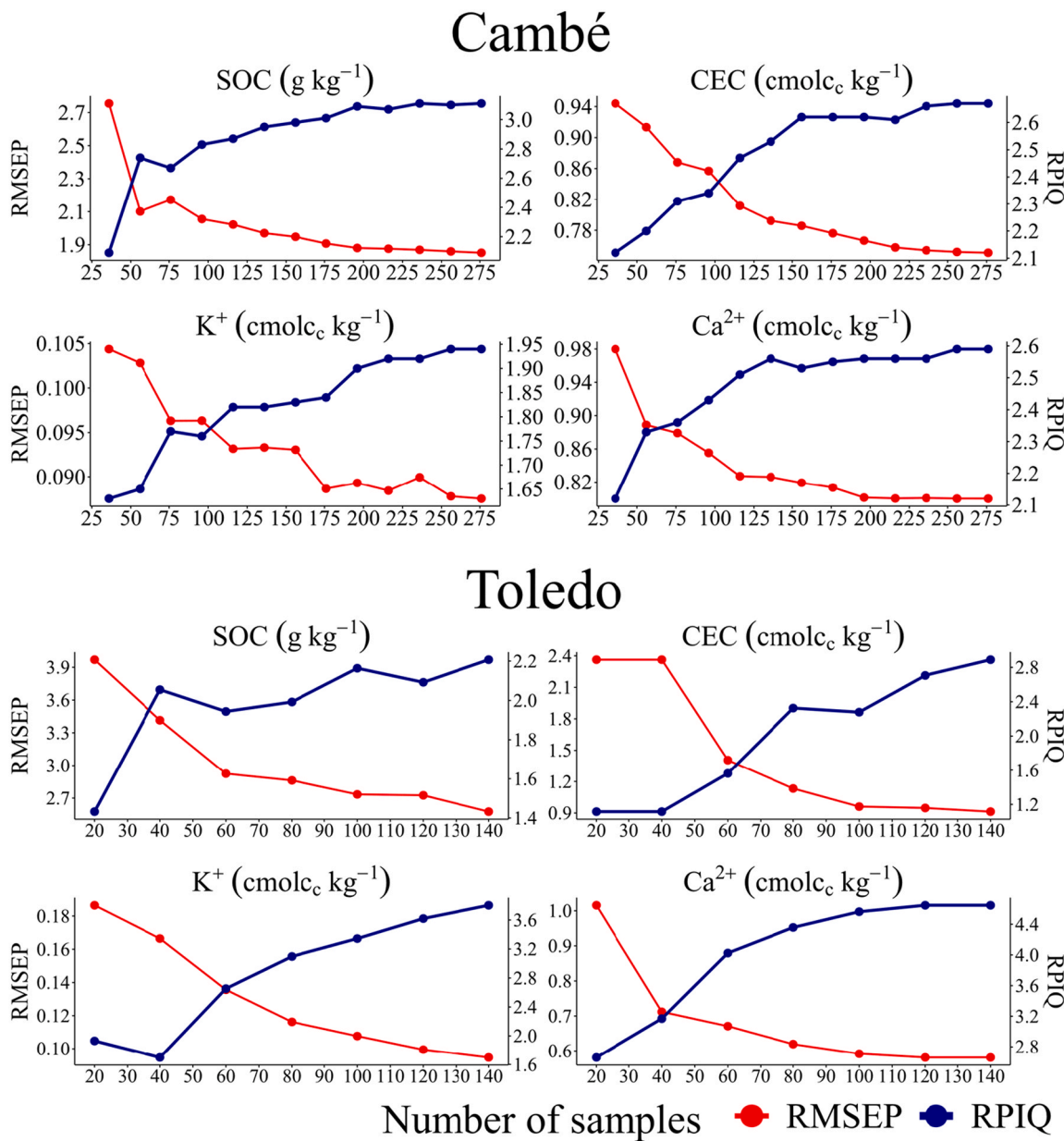


Fig. 4. Prediction results of PLS modeling as a function of the number of calibration samples.

Table 4

Random Test p-values comparing attributes estimated by models calibrated with full and reduced sample sets of Cambé field. Bold values indicated results lower than the p_{critic} of 0.05 for a 95 % confidence level.

Calibration set size	SOC	CEC	K ⁺	Ca ²⁺
256	0.3	0.4	0.3	0.5
236	0.4	0.3	0.052	0.5
216	0.1	0.5	0.3	0.5
196	0.4	0.1	0.2	0.5
176	0.3	0.3	0.3	0.4
156	0.2	0.2	0.06	0.1
136	0.1	0.052	0.03	0.3
116	0.06	0.02	0.04	0.06
96	0.04	0.002	0.001	0.03
76	0.005	0.0	0.02	0.01
56	0.001	0.0	0.004	0.02
36	0.0	0.001	0.03	0.006

Table 5

Random Test p-values comparing attributes estimated by models calibrated with full and reduced sample sets of Toledo field. Bold values indicated results lower than the p_{critic} of 0.05 for a 95 % confidence level.

Calibration set size	SOC	CEC	K ⁺	Ca ²⁺
120	0.3	0.37	0.14	0.9
100	0.2	0.31	0.2	0.2
80	0.2	0.00	0.07	0.052
60	0.09	0.00	0.00	0.01
40	0.01	0.00	0.00	0.02
20	0.00	0.00	0.00	0.00

optimal samples ($n=60$). Furthermore, the highest number of optimal samples of 156 was achieved by the Cambé K⁺ models, which presented the worst performance among all models. Other studies have also reported low performance results for K⁺ predictions (Benedet et al., 2021; Debaene et al., 2014; dos Santos et al., 2021a; Ribeiro et al., 2024). This may be related to the low signal-to-noise ratio of the K line (the most important variable for K⁺ quantification) in pXRF spectra. In this study, the low intensity of the K emission peak (Fig. 2) occurs due to the low concentrations of this element in the soil evaluated.

On the other hand, the optimal calibration set size was shown to be related to the local complexity of the soil matrix, since Cambé — the area with greater soil complexity, *i.e.* Eutrophic Latosol and Nitisol — tended to present an optimal calibration set size with a greater optimal number of samples when compared to Toledo — the area with the simplest matrix, *i.e.* Dystrophic Latosol. Therefore, these results suggested that the optimal calibration set size is also field-specific, besides attribute-specific.

Overall, the present findings align with those of previous studies using local libraries of Vis-NIRS, which showed that model accuracy increases with an increase in the calibration set size (Debaene et al., 2014; Gogé et al., 2014; Lucà et al., 2017). Specifically, Debaene et al. (Debaene et al., 2014) demonstrated through PLS that satisfactory predictions of SOC and Clay could be achieved with as few as 79 calibration samples in Polish soils. Also using PLS, Gogé et al. (Gogé et al., 2014) attained accurate predictions for clay, CEC, CaCO₃, and Fe contents with 94 samples in French soils. Similarly, Lucà et al. (Lucà et al., 2017) found that 115 samples were sufficient for accurate predictions of TC with PLS models in Italian soils. In comparison, the optimal number of samples in our study ranged from 60 to 156, varying with field and soil attributes. Thus, the optimal number of calibration samples employing pXRF-based models were smaller, equivalent or greater than those already reported using Vis-NIR.

3.6. Variables influence on projection (VIP) scores

Fig. 5 presents the VIP scores for models using the full and the

optimal calibration set size scenarios, in which pXRF variables whose associated scores are greater than unity are important for modelling (Mehmood et al., 2012). In general, the most important variables for modeling remained the same when using the full and the optimal calibration set size scenario, with slight fluctuations in their magnitude. These results show the stability in the way PLS uses pXRF spectra to quantify the evaluated soil attributes. In this regard, future studies can use spectral conditions that favor variables with intense VIP scores (*e.g.*, Ca and K emission lines) to obtain even better performances in assessing soil fertility through PLS modeling.

Most of the correlations between soil attributes and spectral data exhibited in the Pearson correlogram (Fig. 3) are reproduced in the VIP scores. For example, Ca- α characteristic X-ray peak was the most important pXRF variable in SOC, CEC, and Ca²⁺ prediction. This was also seen by other studies performed with different tropical Brazilian soils (Andrade et al., 2020; dos Santos et al., 2021a; dos Santos Teixeira et al., 2020; Faria et al., 2020; Ribeiro et al., 2024; Silva et al., 2017; Tavares et al., 2020b). Similar to the Pearson Correlogram, K⁺ models had the K α line with the greatest importance, followed by Ca- α . Although less intense, other pXRF variables also showed the importance in the modeling predictions, namely Al, Si, Ca, Ti, and Fe k-lines. The low mineralogical complexity of tropical soils favors stable relationships between total and available nutrient contents, especially in local libraries. This makes it possible to infer available nutrients from pXRF spectral libraries (Tavares et al., 2023).

3.7. Final considerations

Results indicated that the optimal calibration set size for estimating key soil properties through pXRF libraries and PLS is field- and attribute-specific. In Toledo Field, an area with a Red Dystrophic Latosols, it was suggested an optimal calibration set size of 60 samples for predicting SOC, 80 samples for K⁺ and Ca²⁺ and 100 samples for CEC prediction. While in the Cambé field, an area with soils comprising Red Eutrophic Latosols and Nitisols, 116 samples as optimal calibration set size to predict SOC and Ca²⁺, 136 samples to predict CEC, and 156 for K⁺ prediction. Despite the above-mentioned suggestions, the study also revealed that 5 out of 8 attributes (K⁺ and Ca²⁺ in Toledo, in addition to SOC, CEC, and Ca²⁺ in Cambé) still showed satisfactory predictive performance with RPIQ > 1.70 even with fewer samples in the calibration set ($n=20$ samples in Toledo and $n=36$ in Cambé). These findings indicate that even greater reductions can be considered once the trade-off between performance loss and reduction in calibration set samples is known for the attribute of interest.

Reducing the number of calibration samples offers financial, environmental, and time-saving advantages. Fewer samples reduce traditional analysis expenses, waste generation, and reagent consumption. Additionally, it accelerates the analysis process, which is crucial in situations where quick results are essential.

Finally, it is worth mentioning that this study is the first attempt to address the effect of reducing the calibration set in predictive models for soil fertility attributes when local libraries of pXRF sensors are employed. In light of the findings of this study, to achieve satisfactory predictions simultaneously for all soil attributes addressed (SOC, CEC, K⁺, and Ca²⁺), we suggest maintaining a minimum of about 156 samples in pXRF local libraries considering tropical fields with highly weathered soils.

4. Conclusion

This study responds to the research gap involving the optimization of the calibration set size in local models for predicting fertility attributes using pXRF sensors. Twenty different calibration set size scenarios, comprising two Brazilian agricultural fields, were evaluated. The results demonstrated that it is possible to use a reduced number of samples while maintaining satisfactory predictive performance when applying

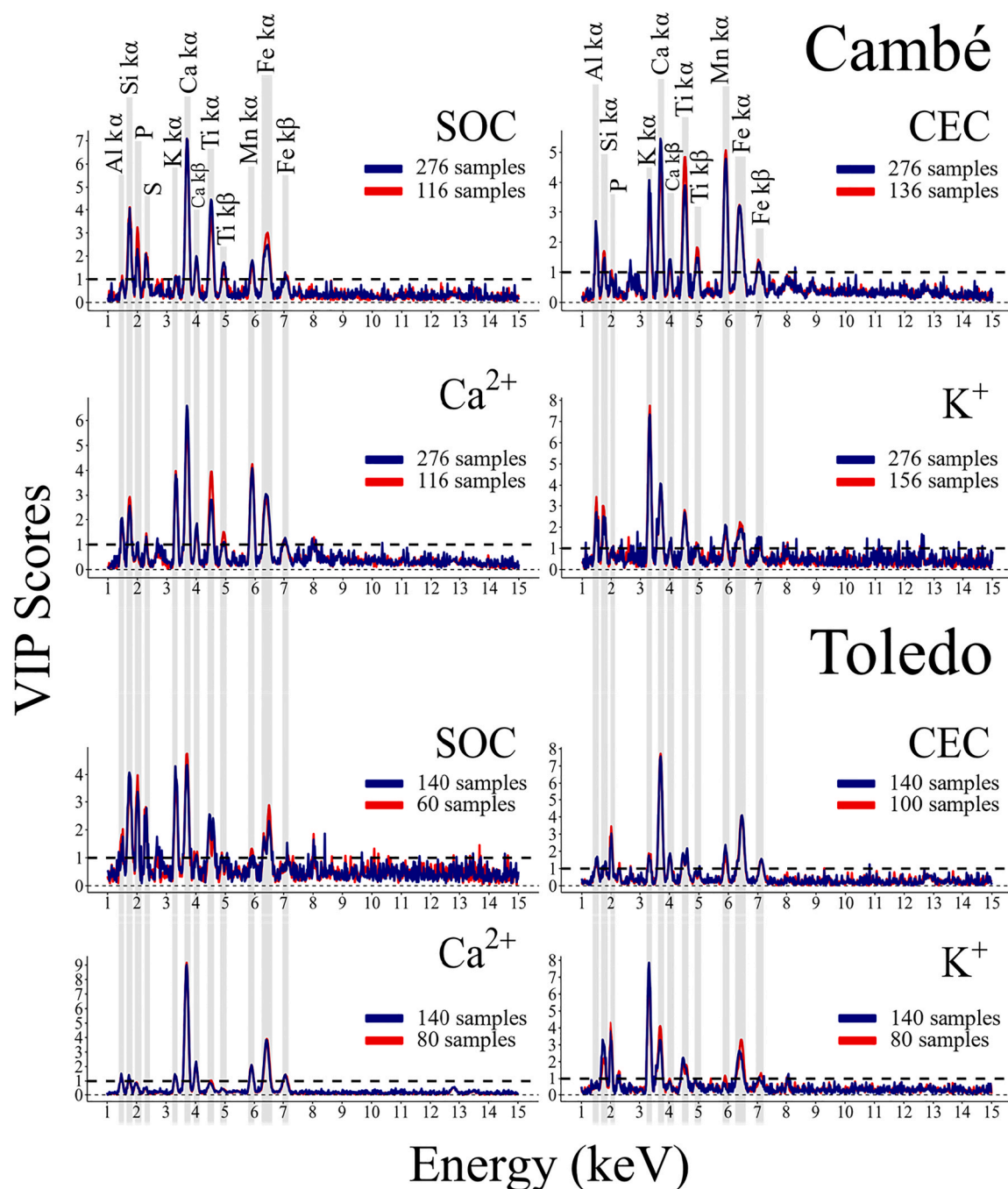


Fig. 5. Variable influence on projection (VIP) scores of the full and optimal reduced calibration models. Values greater than unity (black dotted line) indicate variables that significantly contribute to modeling.

PLS modeling. The optimal number of samples in the calibration set was attribute- and field-specific. For the Cambé library, the minimum number of calibration samples for SOC, CEC, K^+ , and Ca^{2+} models were 116, 136, 156, and 116, respectively. For the Toledo library, 60, 100, 80, and 80 calibration samples were required to calibrate SOC, CEC, Ca^{2+} and K^+ models. Based on RPIQ values, all these models were considered good or very good for predictive quantitative analysis.

The findings can assist users and future researchers to optimize the calibration set size of local libraries. In light of this paper, we suggest around 156 calibration samples as a minimum for local spectral libraries with pXRF sensors in tropical soils. This knowledge is fundamental to avoid extra costs in the application of pXRF as a rapid and environmentally friendly tool for soil fertility analysis in the context of precision

agriculture.

Finally, further studies are encouraged to investigate other soil types and contexts, ML modeling techniques, different sample selection methods, and the temporal stability of models using a reduced calibration set size.

CRedit authorship contribution statement

José Vinícius Ribeiro: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tiago Rodrigues Tavares:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis,

Conceptualization. **José Francirlei de Oliveira:** Writing – review & editing, Validation, Formal analysis. **Graziela M.C. Barbosa:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Fábio Luiz Melquiades:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.soilad.2024.100031](https://doi.org/10.1016/j.soilad.2024.100031).

Data Availability

Data will be made available on request.

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